

# Calculating and Interpreting Means and Standard Errors

**Can Antibiotics Obtained from Soil Bacteria Help Fight Drug-Resistant Bacteria?** Soil bacteria synthesize antibiotics, which they use against species that attack or compete with them. To date, these species have been inaccessible as sources for new medicines because 99% of soil bacteria cannot be grown using standard laboratory techniques. To address this problem, researchers developed a method in which soil bacteria grow in a simulated version of their natural environment; this led to the discovery of a new antibiotic, teixobactin. In this exercise, you'll calculate means and standard errors from an experiment that tested teixobactin's effectiveness against MRSA (methicillin-resistant *Staphylococcus aureus*; see Figure 22.14).

**How the Experiment Was Done** Researchers drilled tiny holes into a small plastic chip and filled the holes with a dilute aqueous solution containing soil bacteria and agar. The dilution had been calibrated so that only one bacterium was likely to grow in each hole. After the agar solidified, the chip was placed in a container containing the original soil; nutrients and other essential materials from the soil diffused into the agar, allowing the bacteria to grow.

After isolating teixobactin from a soil bacterium, researchers performed the following experiment: Mice infected with MRSA were given low (1 mg/kg) or high (5 mg/kg) doses of teixobactin or vancomycin, an existing antibiotic; in the control, mice infected with MRSA were not given an antibiotic. After 26 hours, researchers sampled infected mice and estimated the number of *S. aureus* colonies in each sample. Results were reported on a log scale; note that a decrease of 1.0 on this scale reflects a tenfold decrease in MRSA abundance.

## Data from the Experiment

| Treatment   | Dose (mg/kg) | Log of Number of Colonies | Mean ( $\bar{x}$ ) |
|-------------|--------------|---------------------------|--------------------|
| Control     | —            | 9.0, 9.5, 9.0, 8.9        |                    |
| Vancomycin  | 1.0          | 8.5, 8.4, 8.2             |                    |
|             | 5.0          | 5.3, 5.9, 4.7             |                    |
| Teixobactin | 1.0          | 8.5, 6.0, 8.4, 6.0        |                    |
|             | 5.0          | 3.8, 4.9, 5.2, 4.9        |                    |

## Plastic chip used to grow soil bacteria



**Data from** L. Ling et al. A new antibiotic kills pathogens without detectable resistance, *Nature* 517:455–459 (2015).

## INTERPRET THE DATA

- The mean ( $\bar{x}$ ) of a variable is the sum of the data values divided by the number of observations ( $n$ ):

$$\bar{x} = \frac{\sum x_i}{n}$$

In this formula,  $x_i$  is the value of the  $i$ th observation of the variable (observation number " $i$ " out of a total of  $n$  such observations); the  $\sum$  symbol indicates that the  $n$  values of  $x$  are to be added together. Calculate the mean for each treatment.

- Use your results from question 1 to evaluate the effectiveness of vancomycin and teixobactin.
- The variation found in a set of data can be estimated by the standard deviation,  $s$ :

$$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

Calculate the standard deviation for each treatment.

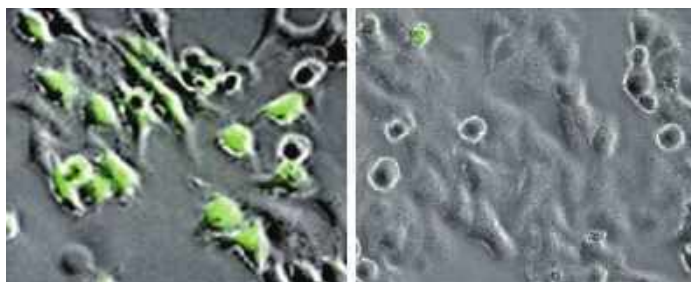
- The standard error ( $SE$ ), which indicates how greatly the mean would likely vary if the experiment was repeated, is calculated as:

$$SE = \frac{s}{\sqrt{n}}$$

As a rough rule of thumb, if an experiment were to be repeated, the new mean typically would lie within two standard errors of the original mean (that is, within the range  $\bar{x} \pm 2SE$ ). Calculate  $\bar{x} \pm 2SE$  for each treatment, determine whether these ranges overlap, and interpret your results.

- ➔ **Instructors:** A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

**▼ Figure 27.24 CRISPR: opening new avenues of research for treating HIV infection.** (a) In laboratory experiments, untreated (control) human cells were susceptible to infection by HIV, the virus that causes AIDS. (b) In contrast, cells treated with a CRISPR-Cas9 system that targets HIV were resistant to viral infection. The CRISPR-Cas9 system was also able to remove HIV proviruses (see Figure 19.8) that had become incorporated into the DNA of human cells.



(a) **Control cells.** The green color indicates infection by HIV.

(b) **Experimental cells.** These cells were treated with a CRISPR-Cas9 system that targets HIV.

each year more than 750 billion pounds of plastic are produced from petroleum and used to make toys, storage containers, soft drink bottles, and many other items. Many of these products end up as plastic waste that contaminates natural habitats and degrades slowly, creating environmental problems (see Concept 56.4). One promising avenue for addressing these problems is to use bacteria that produce natural plastics (Figure 27.25). For example, some bacteria synthesize a type of polymer known as PHA (polyhydroxyalkanoate), which they use to store chemical energy. The PHA can be extracted, formed into pellets, and used to make durable, yet biodegradable, plastics. Researchers are also seeking to reduce the use of petroleum and other fossil fuels by engineering bacteria that can produce ethanol from various forms of biomass, including agricultural waste, switchgrass, and corn.

Another way to harness prokaryotes is **bioremediation**, the use of organisms to remove pollutants from soil, air, or